

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION**CLASS IX • SOCIAL SCIENCE (SUBJECT CODE: 087) • ACADEMIC SESSION 2026-27****SET G — FINAL MASTERY & GRAND CHALLENGE TEST (60 MINS)****CHAPTER 02: SHAPING OF THE EARTH'S SURFACE**

TIME ALLOWED: 60 MINUTES

MAXIMUM MARKS: 25 MARKS

SERIES: CBS-IX-SST-SET-G-02

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

General Instructions (SET G - Final Mastery & Grand Challenge Test):

1. The question paper comprises **25 Olympiad / NTSE / Advanced Analytical Multiple Choice Questions** of **1 mark each**.
2. Designed for **Final Curricular Mastery & In-Depth Analytical Rigour**: Target time is **60 Minutes**.
3. All questions are compulsory. There is no negative marking.
4. Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

**GRAND CHALLENGE SPECIAL:** This final assessment tests your multi-disciplinary analytical reasoning across historical source criticism, geomorphological dynamics, constitutional jurisprudence, and microeconomic market equilibria!

Q1. The Moment Magnitude scale (M_w) quantifies an earthquake's energy release using seismic moment ($M_0 = \mu A D$). What do μ , A , and D represent?

- ☐ A Mass of rocks, Altitude of epicenter, and Distance to station ☐ B Shear modulus of rock (μ), Rupture fault surface area (A), and Average fault slip displacement (D) ☐ C Moisture content, Atmospheric pressure, and Depth of focus ☐ D Magnetic flux, Acceleration of gravity, and Density of crust

Scholarly Concept & Analysis: Seismic moment M_0 equals rock shear rigidity (μ) multiplied by fault rupture surface area (A) and average slip (D).

Q2. An earthquake of magnitude $M_w = 7.0$ releases approximately how many times more seismic energy than an earthquake of magnitude $M_w = 5.0$?

- ☐ A 2 times ☐ B 20 times ☐ C Approximately 1,000 times ($32^2 \approx 1000$) ☐ D 10,000 times

Scholarly Concept & Analysis: Each unit increase on the moment magnitude scale corresponds to a ≈ 31.62 -fold ($10^{1.5}$) jump in energy; two units increase energy by $10^3 = 1,000$ times.

Q3. Why is there an S-wave shadow zone on seismographs extending from 105° to 180° around the Earth from an earthquake epicenter?

- ☐ A S-waves travel too fast to be detected by modern seismographs ☐ B Transverse shear S-waves cannot propagate through the liquid outer core because liquids possess zero shear rigidity ☐ C The Earth's crust is made of soft rubber between 105° and 180° ☐ D S-waves are reflected into outer space by the atmosphere

Scholarly Concept & Analysis: S-waves require shear strength and cannot propagate through the liquid outer core, casting an S-wave shadow beyond 105° .

Q4. In the Mohr-Coulomb equation for mass wasting shear strength ($\tau = c + \sigma_n \tan \phi$), how does water saturation trigger catastrophic landslides?

- ☐ A It freezes the soil into solid ice immediately ☐ B Pore-water pressure reduces effective normal stress ($\sigma' = \sigma_n - u$), drastically lowering shear resistance along failure planes ☐ C Water increases rock cohesion to infinity ☐ D Water stops the force of gravity

Scholarly Concept & Analysis: Elevated pore-water pressure (u) offsets normal overburden stress, reducing frictional shear strength and triggering slope failure.

Q5. On a Hjulström curve diagram, what does the critical relationship between water velocity and sediment particle size reveal?

- ☐ A Boulders erode at lower velocities than fine sand ☐ B Cohesionless medium sand (0.1–0.5 mm) requires the lowest water velocity for initial entrainment, while cohesive clays require much higher velocities to detach ☐ C Water velocity has zero impact on sediment transport ☐ D All sediment sizes deposit at identical water speeds

Scholarly Concept & Analysis: Medium sand entrains easiest; cohesive electrostatic bonds between fine clay particles require surprisingly high velocities to erode.

Q6. In fluvial geomorphology, a river channel is classified as 'meandering' when its Sinuosity Index (Channel Length / Valley Length) is:

- ☐ A Less than 1.0 ☐ B Greater than 1.5 ☐ C Exactly zero ☐ D Negative

Scholarly Concept & Analysis: A stream channel is formally defined as meandering when its sinuosity index (actual channel path length / valley axis length) exceeds 1.5.

Q7. What hydrodynamic mechanism drives the lateral migration of a meandering river loop across its floodplain?

- ☐ A Uniform straight laminar water flow across both banks ☐ B Helicoidal secondary cross-currents causing high velocity and shear erosion on the outer concave cut-bank, while low velocity causes deposition on the inner convex point-bar ☐ C Wind blowing water backwards up the mountain ☐ D Subterranean volcanic eruptions under the riverbed

Scholarly Concept & Analysis: Helicoidal secondary flow accelerates velocity at the outer concave cut-bank (erosion) and decelerates flow on the inner convex point bar (deposition).

Q8. What is the Gutenberg Discontinuity in global geophysics?

- ☐ A The boundary between the troposphere and stratosphere ☐ B The seismic boundary separating the solid silicate mantle from the liquid iron-nickel outer core at a depth of ~2,900 km ☐ C The boundary between continental crust and oceanic crust ☐ D The surface fault line of the San Andreas fault

Scholarly Concept & Analysis: The Gutenberg discontinuity at ~2,900 km depth marks the mantle-core boundary where P-waves decelerate sharply and S-waves terminate.

Q9. In glaciated terrain, what is the 'Bergschrund'?

- ☐ A A small alpine village in Switzerland ☐ B A deep, wide crevasse opening near the headwall of a cirque where active rotational glacial ice pulls away from stagnant ice welded to the bedrock cliff ☐ C A pile of sand dunes formed by wind ☐ D A warm thermal spring melting ice

Scholarly Concept & Analysis: The Bergschrund is the gaping crevasse at the head of a cirque glacier separating moving ice from bedrock-anchored ice.

Q10. Why does chemical dissolution of limestone bedrock in karst terrain proceed faster in cold groundwater than in hot water?

- ☐ A Limestone melts when it freezes ☐ B Carbon dioxide (CO_2) gas is more soluble in cold water, producing higher concentrations of carbonic acid (H_2CO_3), which accelerates calcium carbonate dissolution ☐ C Cold water contains zero oxygen atoms ☐ D Hot water destroys carbonic acid immediately
- Scholarly Concept & Analysis:** Gases dissolve better in cold liquids (Henry's law); cold groundwater holds more dissolved CO_2 , forming stronger carbonic acid.
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Q11. What sharp, knife-edge, serrated rock ridge is formed between two adjacent glaciated cirques eroding toward each other?

- ☐ A Arête ☐ B Drumlin ☐ C Esker ☐ D Alluvial fan

Scholarly Concept & Analysis: An arête is a razor-sharp bedrock ridge separating two adjoining glacial cirques or valleys.

Q12. A steep, pointed, pyramid-shaped mountain peak formed when three or more glacial cirques erode headward toward a central summit is a:

- ☐ A Cirque ☐ B Pyramidal peak (or Horn, e.g., Matterhorn) ☐ C Yardang ☐ D Mushroom rock

Scholarly Concept & Analysis: Glacial horns (pyramidal peaks) form when three or more cirques carve headward into a common mountain crest.

Q13. What streamlined, asymmetrical, tear-drop-shaped hill of glacial till has a steep blunt upstream side and a gentle tapered downstream tail?

- ☐ A Moraine ☐ B Drumlin ☐ C Cirque ☐ D Gorge

Scholarly Concept & Analysis: Drumlins are streamlined, elongated subglacial hills of till shaped by overriding ice, indicating direction of glacier movement.

Q14. Sinuous, winding ridges of stratified sand and gravel deposited by subglacial meltwater streams running inside ice tunnels are:

- ☐ A Moraines ☐ B Eskers ☐ C Barchans ☐ D Yardangs

Scholarly Concept & Analysis: Eskers are long, meandering ridges of glacio-fluvial gravel and sand deposited in subglacial meltwater tunnels.

Q15. Streamlined, wind-carved, knife-edged ridges of rock aligned parallel to prevailing desert winds, separated by scoured furrows, are called:

- ☐ A Mushroom rocks ☐ B Yardangs ☐ C Barchans ☐ D Sinkholes

Scholarly Concept & Analysis: Yardangs are aerodynamic, keel-shaped rock ridges sculpted by wind deflation and abrasion parallel to prevailing winds.

Q16. What coastal landform consists of a narrow ridge of sand or shingle connected to the mainland at one end and terminating in open water at the other?

- ☐ A Sea stack ☐ B Spit (or sand spit) ☐ C Wave-cut platform ☐ D Sea arch

Scholarly Concept & Analysis: A spit is an extended depositional coastal bar formed by longshore drift, projecting out from the coast into an estuary or bay.

Q17. A coastal sand spit that grows completely across a bay mouth, sealing it off from the open ocean and enclosing a shallow lagoon, is a:

- ☐ A Sea stack ☐ B Baymouth bar (or barrier bar) ☐ C Tombolo ☐ D Cirque

Scholarly Concept & Analysis: A baymouth bar completely bridges an embayment, enclosing an isolated brackish coastal lagoon.

Q18. What coastal depositional landform connects an offshore rocky island directly to the mainland with a narrow sand spit?

- ☐ A Tombolo ☐ B Sea arch ☐ C Wave-cut notch ☐ D Alluvial fan

Scholarly Concept & Analysis: A tombolo is a sand bar or spit linking an offshore island to the mainland through wave diffraction deposition.

Q19. What volcanic landform is produced by fluid, low-viscosity, basaltic lava flows that spread over vast distances, creating gentle broad domes?

- ☐ A Steep composite stratovolcano ☐ B Shield volcano (e.g., Mauna Loa) ☐ C Cinder cone ☐ D Acid lava dome

Scholarly Concept & Analysis: Shield volcanoes are broad, low-profile volcanic mountains formed by successive layers of highly fluid basaltic lava.

Q20. Catastrophic collapse of a volcano's summit magma chamber following an explosive eruption creates a gigantic depression called a:

- ☐ A Volcanic crater ☐ B Caldera ☐ C Sinkhole ☐ D Cirque

Scholarly Concept & Analysis: A caldera is a massive volcanic depression (often miles across) formed by the structural collapse of an emptied magma reservoir.

Q21. What is the Mohorovičić (Moho) Discontinuity?

- ☐ A The boundary between the outer core and inner core ☐ B The seismic boundary separating the Earth's crust from the underlying silicate mantle, characterized by a sudden jump in P-wave velocity ☐ C The boundary of the ozone layer in the atmosphere ☐ D The fault plane of a transform plate boundary

Scholarly Concept & Analysis: The Moho marks the crust-mantle boundary where seismic P-waves accelerate from ~6 km/s to ~8 km/s.

Q22. In the chemical weathering of granite, which rock-forming mineral undergoes hydrolysis to transform into soft kaolinite clay?

- ☐ A Quartz ☐ B Potassium feldspar (orthoclase) ☐ C Diamond ☐ D Gold

Scholarly Concept & Analysis: Feldspar reacts with acidic water through hydrolysis ($2\text{KAlSi}_3\text{O}_8 + 2\text{H}_2\text{CO}_3 + 9\text{H}_2\text{O} \rightarrow \text{Al}_2\text{Si}_2\text{O}_5(\text{OH})_4 + 4\text{H}_4\text{SiO}_4 + 2\text{K}^+ + 2\text{HCO}_3^-$), yielding clay.

Q23. The expansion and peeling of curved concentric rock shells from granitic plutons due to tectonic pressure release upon erosion is called:

- ☐ A Frost wedging ☐ B Exfoliation (sheeting) ☐ C Carbonation ☐ D Oxidation

Scholarly Concept & Analysis: Exfoliation occurs when unburdening of deep plutonic rock causes outward expansion along curved pressure-release fractures.

Q24. A semi-circular, bowl-shaped depression at the head of a glacial valley that contains an alpine glacial lake after the ice melts is called a:

- ☐ A Tarn (cirque lake) ☐ B Oxbow lake ☐ C Lagoon ☐ D Sinkhole

Scholarly Concept & Analysis: A tarn is a small mountain lake nestled within an amphitheater-like cirque evacuated by a melted glacier.

Q25. Which plate boundary interaction is responsible for the formation of the volcanic Cascade Range and the Andes Mountains?

- ☐ A Divergent oceanic rifting ☐ B Subduction of an oceanic plate beneath a continental plate at a convergent boundary ☐ C Transform lateral strike-slip faulting ☐ D Continental-continental collision without subduction

Scholarly Concept & Analysis: Oceanic-continental convergence forces the dense oceanic plate to subduct, generating magma that builds volcanic cordilleras.

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET G)

01 A B C D	02 A B C D	03 A B C D	04 A B C D	05 A B C D
06 A B C D	07 A B C D	08 A B C D	09 A B C D	10 A B C D
11 A B C D	12 A B C D	13 A B C D	14 A B C D	15 A B C D
16 A B C D	17 A B C D	18 A B C D	19 A B C D	20 A B C D
21 A B C D	22 A B C D	23 A B C D	24 A B C D	25 A B C D

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & IN-DEPTH SCHOLARLY EXPLANATIONS (SET G)

Q. No.	Correct	Curricular Concept & Scholarly Analysis
1	B	Seismic moment M_0 equals rock shear rigidity (μ) multiplied by fault rupture surface area (A) and average slip (D).
2	C	Each unit increase on the moment magnitude scale corresponds to a ≈ 31.62 -fold ($10^{1.5}$) jump in energy; two units increase energy by $10^3 = 1,000$ times.
3	B	S-waves require shear strength and cannot propagate through the liquid outer core, casting an S-wave shadow beyond 105° .
4	B	Elevated pore-water pressure (u) offsets normal overburden stress, reducing frictional shear strength and triggering slope failure.
5	B	Medium sand entrains easiest; cohesive electrostatic bonds between fine clay particles require surprisingly high velocities to erode.
6	B	A stream channel is formally defined as meandering when its sinuosity index (actual channel path length / valley axis length) exceeds 1.5.
7	B	Helicoidal secondary flow accelerates velocity at the outer concave cut-bank (erosion) and decelerates flow on the inner convex point bar (deposition).
8	B	The Gutenberg discontinuity at $\sim 2,900$ km depth marks the mantle-core boundary where P-waves decelerate sharply and S-waves terminate.
9	B	The Bergschrund is the gaping crevasse at the head of a cirque glacier separating moving ice from bedrock-anchored ice.
10	B	Gases dissolve better in cold liquids (Henry's law); cold groundwater holds more dissolved CO_2 , forming stronger carbonic acid.
11	A	An arête is a razor-sharp bedrock ridge separating two adjoining glacial cirques or valleys.
12	B	Glacial horns (pyramidal peaks) form when three or more cirques carve headward into a common mountain crest.
13	B	Drumlins are streamlined, elongated subglacial hills of till shaped by overriding ice, indicating direction of glacier movement.
14	B	Eskers are long, meandering ridges of glacio-fluvial gravel and sand deposited in subglacial meltwater tunnels.
15	B	Yardangs are aerodynamic, keel-shaped rock ridges sculpted by wind deflation and abrasion parallel to prevailing winds.
16	B	A spit is an extended depositional coastal bar formed by longshore drift, projecting out from the coast into an estuary or bay.
17	B	A baymouth bar completely bridges an embayment, enclosing an isolated brackish coastal lagoon.
18	A	A tombolo is a sand bar or spit linking an offshore island to the mainland through wave diffraction deposition.
19	B	Shield volcanoes are broad, low-profile volcanic mountains formed by successive layers of highly fluid basaltic lava.
20	B	A caldera is a massive volcanic depression (often miles across) formed by the structural collapse of an emptied magma reservoir.
21	B	The Moho marks the crust-mantle boundary where seismic P-waves accelerate from ~ 6 km/s to ~ 8 km/s.

Q. No.	Correct	Curricular Concept & Scholarly Analysis
22	B	Feldspar reacts with acidic water through hydrolysis ($2\text{KAlSi}_3\text{O}_8 + 2\text{H}_2\text{CO}_3 + 9\text{H}_2\text{O} \rightarrow \text{Al}_2\text{Si}_2\text{O}_5(\text{OH})_4 + 4\text{H}_4\text{SiO}_4 + 2\text{K}^+$), yielding clay.
23	B	Exfoliation occurs when unburdening of deep plutonic rock causes outward expansion along curved pressure-release fractures.
24	A	A tarn is a small mountain lake nestled within an amphitheater-like cirque evacuated by a melted glacier.
25	B	Oceanic-continental convergence forces the dense oceanic plate to subduct, generating magma that builds volcanic cordilleras.